

CS224R Milestone: IPO + RLOO Alignment Report

Project Title: Your Project Title Here

Team Members: Member 1, Member 2, Member 3

Emails: email1@stanford.edu, email2@stanford.edu, email3@stanford.edu

Submission Checklist & Instructions (Please remove this block before submitting):

1. **IPO Milestone Assignment (Gradescope):** Submit `ipo_trainer/ipo.py` (implementation) and `eval.json` (evaluation output).
2. **RLOO Milestone Assignment (Gradescope):** Submit `rloo_trainer/rloo_update_worker.py` (implementation) and `eval.json` (evaluation output).
3. **IPO/RLOO Milestone Submission PDF (Gradescope):** Submit this compiled PDF report containing all required plots and rollout samples detailed below.
4. **Plots Requirement:** All reported training and evaluation metrics must be displayed as plots (e.g., generated with `matplotlib`) rather than only scalar values.

1 IPO Milestone Component

1.1 Training and Evaluation Metrics

Describe your IPO implementation, training setup (e.g., hyperparameter choices such as learning rate, β , epochs, optimizer), and any specific observations during the alignment process.

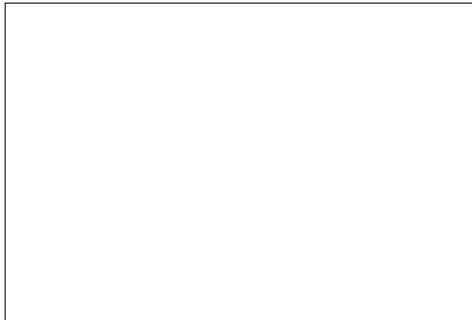


Figure 1: IPO Loss on Train and Test sets over training iterations.

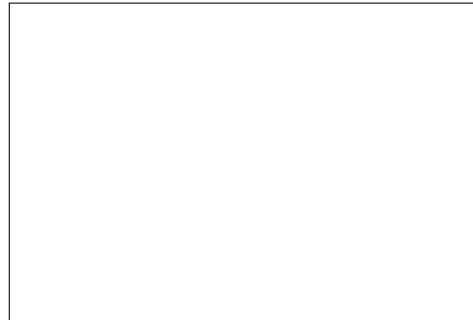


Figure 2: IPO Reward Margin on Train and Test sets over training iterations.

1.1.1 Functional Correctness ($\text{Pass}@k$)

Report the $\text{pass}@k$ metrics (e.g., as a bar plot generated with `matplotlib` comparing SFT and IPO on the Test Set).



Figure 3: $\text{Pass}@k$ comparison between SFT Baseline and IPO Aligned Model on the Test Set.

1.2 Sample Rollout

Include one test-set rollout for a sample test problem using the IPO-aligned model.

```
[PROMPT]
Target: 24
Allowed numbers: [3, 4, 6, 8]

[IPO MODEL ROLLOUT]
<think>
[model thinking here]
</think>
<answer>[model answer here]</answer>
```

Listing 1: Sample IPO Test-Set Rollout (Countdown)

2 RLOO Milestone Component

2.1 Training and Evaluation Metrics

Describe your RLOO implementation (such as the REINFORCE Leave-One-Out baseline configuration), RL training setup (e.g., learning rate, KL coefficient, number of rollouts per prompt), and stability observations.



Figure 4: RLOO Loss on Train set over training iterations.

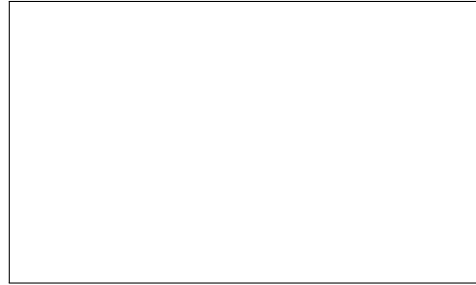


Figure 5: Importance Weight Mean on Train set.

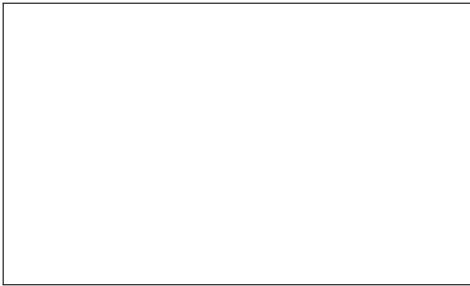


Figure 6: KL Loss on Train set over training iterations.

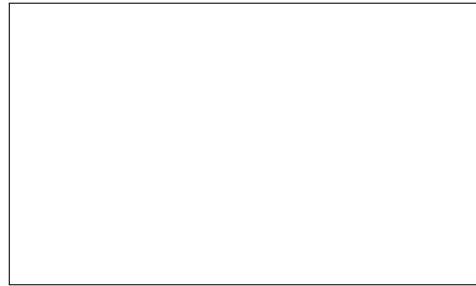


Figure 7: Rollout Accuracy (over 16 samples) on Train set.

2.1.1 Functional Correctness (Pass@ k)

Report the pass@ k metrics (e.g., as a bar plot generated with matplotlib comparing SFT and RLOO on the Test Set).



Figure 8: Pass@ k comparison between SFT Baseline and RLOO Aligned Model on the Test Set.

2.2 Sample Rollout

Include one test-set rollout for a sample test problem using the RLOO-aligned model.

[PROMPT]
Target: 24

```
Allowed numbers: [3, 4, 6, 8]

[RLOO MODEL ROLLOUT]
<think>
[model thinking here]
</think>
<answer>[model answer here]</answer>
```

Listing 2: Sample RLOO Test-Set Rollout (Countdown)

3 Extension Updates (Optional)

Describe any progress, modifications, or updates regarding your project extension here. (You may remove or leave this section blank if you do not have updates to report at this stage).

References